Funded by the European Union

Application Form - E-Sailors

Electric solar wind Sail doctors

Doctoral Network (DN)
Marie Skłodowska-Curie Action
Call: HORIZON-MSCA-2024-DN-01

1. PERSONAL DATA

First Name	Reza
Family Name	Aliasgari Renani
Email Address	rezaaliasgarirenani@gmail.com
Postal Address	141707, Moscow Oblast, Dolgoprudny, Pervomayskaya Street, Building 32, Block 3, Apartment 381-2
Country of Citizenship	Iran
Nationality	Iranian
Gender	Male

2. COUNTRY OF RESIDENCE and COUNTRY OF MAIN ACTIVITY DURING THE LAST 3 YEARS

Please indicate your country of residence and country where you carried out main activity (study, work, etc.) in the last 3 years. This information is needed to determine if applicants fulfill the EU Mobility Rule for the position(s) applied for- *Researchers must not have resided or carried out their main activity (e.g., work, studies) in the country of the recruiting beneficiary for more than 12 months in the 36 months immediately before their recruitment date.*

Starting date	**End date**	**Country**
11/01/2023	Present	Russian Federation

3. EDUCATION AND QUALIFICATIONS

Please indicate your degree qualifications below.

BACHELOR'S DEGREE

Institution/university	Moscow Institute of Physics and Technology (National Research University)
City, country	Dolgoprudny, Moscow Oblast, Russian Federation
Major field of study	Technical Physics
Start and end dates	09/2020 - 06/2024

MASTER'S DEGREE

Institution/university	Moscow Institute of Physics and Technology (National Research University)
City, country	Dolgoprudny, Moscow Oblast, Russian Federation
Major field of study	Applied Mathematics and Physics
Master's degree final mark	4.70/5.00 (Currently studying, this mark is for the 1st year.)
Final thesis title	Investigation of radiation-induced effects on programmable microelectronic systems by means of plasma-based facilities simulating space flight conditions.
Thesis supervisor(s)	Mikhail Vasiliev, Viacheslav Chesnokov
Start and end dates	09/2024 - 06/2026
Number of ECTS obtained	120
GPA	4.7/5.00

4. RECORD OF PROFESSIONAL EXPERIENCE

Professional experiences relevant to the research position you are applying for (specify up to three experiences in chronological order, starting from the most recent):

- **Design Center for the Development of Microprocessor Technology for AI Systems, SoC Development Laboratory (MIPT)**
  - **Role**: Programmer / RTL Design Engineer
  - **Dates**: Sep 2024 -- Present
  - **Location**: Moscow, Russian Federation
  - **Tasks**: Verilog DSP Implementations, HDL Code Generation, FPGA verification and synthesis, Radiation/plasma testing of FPGA devices

- **Institute of Microelectronics Technology, Russian Academy of Sciences (IMT, RAS)**
  - **Role**: Laboratory Researcher
  - **Dates**: Mar 2023 -- Aug 2024
  - **Location**: Moscow, Russian Federation
  - **Tasks**: Installed and automated measurement setups, Semiconductor device characterization and analysis, Data processing and model comparison

- **Laboratory of Modeling of Mechanical Systems and Processes (MIPT)**
  - **Role**: Engineer / Technician
  - **Dates**: Mar 2023 -- Aug 2024
  - **Location**: Moscow, Russian Federation
  - **Tasks**: Design of CubeSat deployer fixtures in SolidWorks, Vibration simulation and testing

5. ADDITIONAL SKILLS

Language skills (language and level):

- English: C2 (TOEFL iBT 113)
- Russian: B1
- German: B1 (ÖSD)
- Farsi: Native

Relevant computer skills (software, programming, etc. and specify user level: basic, average, experienced):

- **Advanced:**
  - Automation of experimental techniques
  - Electrical characterization
  - Data processing, LaTeX
  - MATLAB, Simulink, HDL Coder
  - Verilog, RTL design
  - FPGA development
  - Fixed-point computations
  - SciPy, Cocotb, OpenCV

- **Intermediate:**
  - Python, C++, Arduino
  - Vivado, Vitis
  - Git, Unix/Linux OS
  - OriginLab
  - SolidWorks
  - ERDAS IMAGINE
  - PCB, EasyEDA
  - OpenRocket

Please specify any relevant professional teamwork experience (and your role within the team):

In all of my research roles I worked in teams. At the Russian Academy of Sciences, I worked under two supervisors who defined the research questions and experimental methods. My role was to implement experiment automation (MATLAB), run measurements with them, process the data, and prepare plots and summaries with their help and guidance. I translated their theoretical requirements into practical procedures and refined the setup together. At the SoC Development Lab, our team divided the image-signal-processing pipeline among several engineers. I was responsible for Simulink-to-HDL flow, generated and validated HDL, and compared it against my own and colleagues' hand-written Verilog after Vivado synthesis. We co-developed testbenches and verification and coordinated through GitLab.

6. STATEMENT AND SIGNATURE

The data you provide will be used to process your application to the E-Sailors Consortium. The data will be kept for a period of five years after the end of the project for the purpose of an audit by the EU.

Printed name: REZA ALIASGARI RENANI

Location: Dolgoprudny, Moscow Oblast, Russian Federation

Date: 08/01/2026

Signature: